How Tire Degradation Impacts Formula 1

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2/15/2025

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Tire degradation is a crucial aspect of the sport of Formula 1 (F1) that provides the ultimate technical challenge that can win or lose a race. As F1 is the premiere racing series, it also serves as a cutting-edge industry for automotive technology. The tires used in this series are specially manufactured and supplied by Pirelli to be able to withstand the high cornering forces and intense speed. But in order to maximize performance, the tires only perform optimally within a specific temperature window. This degradation is what has led to a better showing of races but also has made this topic an on-going extensive study for teams and researchers.

Tire degradation is a significant aspect to the F1 community as it directly impacts race outcomes, driver safety, and team strategies. This has resulted in the need for *tire management,* the prevention of loss of optimal tire performance over time due to factors like heat, friction, and road conditions. This includes pit stop strategy of when to change tires during the race, which tire compound to use, and what driving style needs to be employed. Recent development within F1 like the introduction of new tires each year and changes in race regulations have heightened the focus on degradation and the need to gather data constantly. But as more research is conducted and more data is analyzed, teams can more accurately predict and manage how a tire degrades under certain circumstances and situations.

This paper aims to investigate how do F1 teams integrate data analysis and technological advancements to optimize the tire performance and inform race strategies. To address this, I will analyze how and why tire degradation is measured and how teams and drivers respond to the challenges of tire management. From a wider global perspective, advancements in tire technology developed for F1 often translate to improvements in consumer vehicles so the case studies, real race results, and research analysis can better the lives of the public including an average non-racing fan.

The Technicality of Tire Degradation

Tire degradation is used to signal the end of a tire's life during the race and comes because of the heat and pressure while racing. This leads to a narrow specific temperature window being the most ideal for peak tire performance before degradation causes performance to fall off. To understand why this phenomenon happens, an understanding of how heat is generated and transferred in the tire is essential. According to Holloway (2019), an owner of racing technology specifically used for tire tracking, tires can gain heat by being generated through hysteresis- friction from road contact and conductive heat transfer from the wheel. Conversely, heat can be lost convectively to the surrounding air, conductively to the road, and radiatively. There needs to be a balance between the heat gain and loss to keep within the temperature window. Too little heat can also cause tires to be cold and unresponsive while too hot of tires shorten the lifespan, causing tire degradation. But a tire will eventually succumb to the amount of heat being applied as it is not able to output enough heat to offset the cycle. A tire's grip is completely generated by indentation and molecular adhesion, two temperature-dependent factors (Holloway, 2019). Indentation refers to the tire's compression and relaxation cycle as it goes over the track's surface allowing the tire to be able to mold and grab onto the track. Molecular adhesion refers to the Van der Waals bonding that adheres the rubber molecules in the tire to the roughness of the road surface. These two factors work in tandem to keep the wheels gripped to the track with friction and are essential to be able to make turns under such high speeds with minimal braking. This explains why constant simulations are ran on tire wear before a race time to be understanding of how the heat will affect the tire's ware as it varies from track to track with different surfaces. Simulations can give clear models to most situations and how to respond properly. Data evidence is also able to conclude how on what specific part of a circuit and when heat is too quickly being generated unnecessary, overloading the tires.

A tire's ware and degradation can cause the reduction of grip and tread-wear (Westa & Limebeer, 2022). Tire degradation can present itself in many ways visible including blistering on the tire surface that leads to the tires sliding across the track surface. It is comparable to how wood is worn away by sandpaper. Blistering of the tire refers to the inner part of the tire over heating that shows bubbles on the tire surface compared to graining which is heat build-up where small rubber chunks break off sticking to track. These effects shorten the lifespan of a tire causing it to be replaced or being at risk of exploding. The research presented by Westa & Limebeer (2022), respected engineers in the field of motorsports, concludes that the first lap on a new set of tires is a relatively slow lap due to still being cold, by the third lap the temperature was in optimal operating range, and by lap 10 the lap times were now like the first lap. This asserts that a lap stint will be best in the middle and declines in the beginning and end of a stint. In an overall view of 40-70 lap, races depending on the circuit, would require precision and attention to push a tire's limit for as long as possible. Another factor to consider about tire degradation is the difference between each tire within a set. A tire gains and loses heat differently based on what position they are used like left front, right front, left rear, and right rear. This also affects how quickly they would get warmed and the time it would take to start graining or blistering. F1 is a motor sport specific to circuits compared to other motorsport series that also race oval tracks. This means F1 has tracks that contain a variety of left turns and right turns that will affect each tire differently on a singular car. Thus, explains why constant simulations of tire-wear models are ran so teams can understand when a tire is degrading too much and becoming slower.

How Tire Degradation Affects Race Strategy

As tire degradation is a central variable in competitive lap times, it plays a significant role in race strategy within F1. The major thing is that tire wear will dictate pit stops. Tire degradation forces teams to plan pit stops as tires can only function for a certain number of laps as shown during prior race simulations. As noted by De Groote (2021), a researcher within the engineering department at Delft University of Technology, pitstops can add around 20 seconds to a lap in a sport where margins of winning are usually less than five seconds. But this also notes that pitstops not only serve to replace worn tires but can also be used to regain lost track positions. Newer fresher tires, free from degrading, perform better and can save time while the other cars on track still are on worn tires. On the other hand, teams can choose to preserve tires by not pushing so hard for longer lap stints which will mean that driver will end the race with faster lap times and fresher tires. The data shown in past races and simulation analysis also helps determine whether teams will choose to take one or two pitstops. Two pitstops would take more time to accomplish but if a track supports high degradation, it would be necessary to offset the slowness of a blistered tire seen during one pit-stop strategies. Low degradation tracks would support single stops to avoid additional time wasted in stops. High degradation tracks are considered typically hotter circuits that add even more heat to the tires or tracks that have a rougher surface like street-circuits where roads are not specifically made for racing. Even though F1 is a sport of racing, it is sometimes able to be won off track during pit strategy.

The study by Rockerbie & Easton (2022), researchers from University of Lethbridge and Simon Fraser University respectively, further emphasizes the interaction between how a driver's competitive balance can efficiently manage tire wear in accordance with a specific circuit. Circuit specifically like Australia and Miami has teams opting to use either a one-stop or two-stop strategy based on how a driver uses his tires. A driver with more conservative driving can decide not to waste time with an extra pitstop and can adopt a one-stop strategy that also leads to a slower race pace due to managing the degrading of the same set of tires. But a driver who is confident with overtaking and charging through the field can decide on a two-stop race strategy and can push harder to degrade but might have difficulty overtaking. Ultimately, it comes down to how a team decides to handle strategies based on the race simulations run prior to each race that provides a set of data on how the tires will fare under each strategy and how to adopt the strategy compared to the other teams on track. Tire management, although very researched, still needs a lot of intuition for it and is evolving as more teams gather more data.

Newey (2017), a renowned race engineer, has commented in his autobiography how the 2005 season saw a shift in regulations that disadvantaged Bridgestone which had excellent tires to manage degradation but worse for top speeds compared to Michelin, whose fast tires purposely degraded. This adds context to how just less than 20 years ago F1 as a sport used different tire manufacturers. Pirelli, who is currently the sole supplier of F1 tires, did not enter the sport until 2011. Every year, new tires are developed based on more data and new regulations cause new simulations needing to be run and knowledge gathered just one season ago is considered outdated. As the developed for the year is used, the degradation on it can be studied as the season passes as it is not a black or white issue but flexible due to many factors affecting the degradation of a set of tires.

Teams and drivers specifically, must be able to balance pushing a car and tires harder for better lap times versus preserving tire life to reduce pit stops. This difference can be seen with more aggressive driving styles characterized by late braking, high cornering loads, rapid acceleration that generates heat quicker which burns through the rubber quicker as well versus a smoother driving style that prioritizes race consistency over pushing the car to its limits every lap. The study conducted by de Winter & de Groot (2012), "The ability of a racing driver to control a car's braking force has a major effect on lap times." This is true as drivers like Lewis Hamilton who extend tire life by famously controlling their braking while maintaining strong pace have succeeded in being one of the best drivers ever seen in F1. He can control the slip and temperature buildup, allowing them to take advantage of fewer pit stops and gain strategic advantages. The real time telemetry allows teams to see in real-time specifically what corners a driver breaks too heavy by comparing the data to the one gathered during the simulations prior to the race start. This specificity has been developed over time so that a singular lap-time can be broken down even more in sectors and corners providing detailed feedback to the drivers.

Conclusion

Tire degradation plays a pivotal role in Formula 1, influencing race outcomes, team strategies, and the overall evolution of motor sport technology. The balance between optimizing performance and managing degradation is an ongoing battle for the teams that require extensive data analysis, simulations, and strategic decision-making. Ultimately, it's about the teams and drivers fighting against the natural tire evolution of degradation to cross the finish line first.

Furthermore, tire management extends beyond motorsports and the industry of F1, but the research and innovations used affect day-to-day consumer vehicles. The data analyzed will refine F1's teams' approach to ensuring races remain competitive, strategic, and safe but also will improve the tire health for everyday drivers to remain safe and efficient. The study of tire degradation is not about extending tire life, but about maximizing performance and maintaining control, a delicate problem that can define success in Formula 1.

References

de Groote, J. (2021). Overtaking in Formula 1 during the Pirelli era: A driver-level analysis. Journal of Sports Analytics, 7(2), 119–137. https://doi.org/10.3233/JSA-200466

de Winter, J. C., & de Groot, S. (2012). The effects of control-display gain on performance of race car drivers in an isometric braking task. Journal of Sports Sciences, 30(16), 1747–1756. https://doi.org/10.1080/02640414.2012.713978

Holloway, C. (Year). Influence of temperature on tire grip [White paper]. Izze Racing.

Rockerbie, D. W., & Easton, S. T. (2022a). Race to the podium: Separating and conjoining the car and driver in F1 Racing. Applied Economics, 54(54), 6272–6285

Newey, A. (2017b). How to Build a Car: The Autobiography of the World's Greatest Formula 1 Designer. HarperCollins UK.

West, W. J., & Limebeer, D. J. N. (2020b). Optimal tyre management for a high-performance race car. Vehicle System Dynamics, 60(1), 1–19. https://doi.org/10.1080/00423114.2020.1802047